

OBJECTIVE / SUMMARY

A passionate software engineer with 4+ years of hands-on experience building scalable backend systems and data pipelines. I am seeking a role where i can leverage my skills in Python, cloud infra, and ML engineering. Have worked in fast-paced startups and mid-size companies — comfortable with ambiguity and shifting priorities. Interested in NLP, LLMs, and building real products that people actually use (not just demos).

WORK EXPERIENCE

Backend Engineer — AI Products

Jan 2022 – Present

Instabug, Cairo (Remote-hybrid)

- Designed and deployed a document intelligence pipeline using FastAPI + ChromaDB that reduced manual ticket triage time by ~40%.
- Built a multi-tenant RAG system ingesting 10k+ support PDFs and Notion docs; used sentence-transformers (all-MiniLM-L6-v2) for embeddings.
- Owned the Docker / docker-compose setup for all AI microservices; reduced cold-start time by 60% through layer caching and model preloading.
- Wrote custom PDF parsers for messy scanned documents (Arabic + English mixed), handling RTL text extraction edge cases.
- Collaborated with product team to ship an internal LLM-powered search feature used by 200+ support agents daily.

Junior Python Developer

Aug 2020 – Dec 2021

Synapse Analytics, Cairo

- Developed ETL pipelines using Apache Airflow to ingest and transform structured + semi-structured data from client ERP systems.
- Maintained a Django REST API serving 3 enterprise clients; improved query performance by 35% via index optimization and caching (Redis).
- Onboarded 2 junior developers and led weekly code review sessions.
- Technologies: Python 3.8, PostgreSQL, Redis, Celery, Docker.

Freelance ML/Data Projects

2019 – 2020 | Various clients via Upwork & direct referrals

Completed 12+ freelance projects ranging from sentiment analysis on Arabic social media text, to building simple recommendation engines for e-commerce clients. Worked mostly solo, managed own deadlines. Some projects used scikit-learn, others used basic fine-tuning of BERT models (HuggingFace). Not all projects are listed here due to NDAs.

EDUCATION

B.Sc. Computer Science — Ain Shams University, Cairo

Graduated: June 2020

GPA: 3.4 / 4.0 | Graduation Project: Arabic Sentiment Analysis using BiLSTM (Grade: A)

Relevant Coursework: Natural Language Processing, Machine Learning, Distributed Systems, Algorithms & Data Structures, Database Systems, Software Engineering.

TECHNICAL SKILLS

Languages: Python (expert), SQL, Bash, some JavaScript (not my strong suit)

Frameworks: FastAPI, Django, Flask, Celery, LangChain (basic)

ML / NLP: scikit-learn, HuggingFace Transformers, sentence-transformers, FAISS, ChromaDB, NLTK

Infra / DevOps: Docker, docker-compose, GitHub Actions, AWS (EC2, S3, RDS), basic Kubernetes

Databases: PostgreSQL, Redis, MongoDB, ChromaDB, Qdrant (explored)

Other: Git, Linux, REST API design, MVC architecture, Agile/Scrum

CERTIFICATIONS & COURSES

- AWS Certified Developer – Associate (2023)
- Deep Learning Specialization — Coursera / deeplearning.ai (2021)
- Hugging Face NLP Course — Certificate of Completion (2022)

SELECTED PROJECTS

Arabic CV Matcher (RAG System) (2023)

End-to-end RAG pipeline that accepts a job description and ranks uploaded CVs by relevance. Used PyMuPDF for Arabic PDF parsing, CAMEL Tools for morphological normalization, and Qdrant as vector DB. Deployed via docker-compose. Achieved ~78% precision@5 on a manually labeled test set of 50 JD-CV pairs.

LLM Factory API (2023)

A FastAPI service implementing the Factory Design Pattern to abstract over multiple LLM backends (OpenAI GPT-4, Gemini 1.5, local Ollama/Mistral). Switching models requires changing one environment variable. Used by 3 internal teams at Instabug for prototyping.

Airbnb Review Sentiment Dashboard (2021)

Scraped ~50k Cairo Airbnb reviews, ran BERT-based sentiment analysis, and built a Streamlit dashboard. Found that cleanliness complaints spike in summer months. Published as a Medium article (2.1k views).

LANGUAGES & OTHER

Arabic (native) | English (fluent, C1) | French (basic, A2)

References available on request. Open to relocation for the right opportunity. GitHub: github.com/ahmedkarim-dev (some repos private)